

FLUXLOCK WHITEPAPER v2

Continuity-Based Adaptive Identity Protocol Protocol Specification Edition Author: Roger Pelletier

Abstract

Fluxlock is a continuity-based identity protocol designed to replace static authentication systems with persistent, verifiable continuity. Traditional identity systems rely on possession of passwords, cryptographic keys, certificates, or stake ownership. Fluxlock introduces a different model: Identity is Continuity.

1. The Identity Problem

Modern authentication systems are structurally fragile because they rely upon static trust assumptions. Possession = Identity creates compromise risk, trust discontinuity, and governance centralization.

2. Identity as Continuity

Identity becomes the persistent continuity of an entity across time. Authentication becomes evaluation of continuity rather than verification of possession.

3. Core Protocol Architecture

Identity Layer, Continuity Layer, Memory Layer, Governance Layer, Authority Layer, and Consensus Layer work together to preserve trust through evolution.

4. Continuity Memory

Validators maintain persistent memory including stable epochs, fractures, recoveries, inherited trust, governance history, and historical stability.

5. Continuity States

Healthy, Evolving, Recovering, Rehabilitating, Quarantined, Fractured, and Exiled states determine participation and legitimacy.

6. Distributed Continuity Mesh

Validators influence surrounding nodes through trust gravity, stabilization fields, healing waves, resonance fields, entropy propagation, and topology effects.

7. Identity Lineage

Every identity maintains a lineage chain recording continuity-preserving evolution. Identity may change. Continuity remains.

8. Authority Continuity

Authority emerges from demonstrated continuity, reputation, memory, stability, and consensus performance.

9. Authority Elections

Leadership emerges from proven continuity through peer evaluation of trust, memory, reputation, and historical performance.

10. Authority Legitimacy

Authority holders maintain legitimacy scores influenced by rehabilitation history, continuity state, fractures, and governance behavior.

11. Authority Challenges

Low-legitimacy leaders may be challenged by validators with stronger continuity metrics, preserving accountability.

12. Authority Succession

Leadership continuity can be inherited through verified continuity lineage while preserving institutional trust.

13. Post-Quantum Foundations

Post-quantum cryptography secures continuity proofs, validator identities, lineage chains, and succession verification.

14. Emergent Network Behavior

Trust clustering, resonance amplification, fracture containment, stabilization zones, and continuity reinforcement emerge at scale.

15. Security Model

Fluxlock assumes compromise will occur and focuses on detection, containment, rehabilitation, and continuity preservation.

16. Future Research

Regional continuity clusters, trust weather systems, adaptive topology evolution, and self-healing governance networks.

Conclusion

Traditional systems ask: Who possesses the credential? Fluxlock asks: Has continuity been preserved? Fluxlock is a continuity-based identity protocol.